

Problem K. K-MEX

Time Limit: 1 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

Let a be an array of length n . Define $\text{mex}(a)$ as the smallest non-negative integer that does not appear in a . For example:

- $\text{mex}([4, 0, 2, 1]) = 3$
- $\text{mex}([-2, -4, 3]) = 0$

Define a k -transformation as follows: for each index i ($1 \leq i \leq n$), independently choose to either keep a_i unchanged or replace it with $k - a_i$. After all choices are made, we obtain a new array a' .

Define $k\text{-mex}(a)$ as the maximum possible value of $\text{mex}(a')$ among all arrays a' that can be obtained from a via a k -transformation.

You must answer q queries. Each query provides a non-negative integer k ; compute $k\text{-mex}(a)$ for each query.

To reduce the output size, you are only required to output the bitwise XOR (exclusive-OR) of the answers over all q queries.

Input

The first line contains a single integer T ($1 \leq T \leq 10^3$) --- the number of test cases.

For each test case:

- The first line contains a single positive integer n ($1 \leq n \leq 5 \times 10^3$), denoting the length of the array.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$), denoting the initial elements of array a .
- The next line contains a single positive integer q ($1 \leq q \leq 5 \times 10^5$), denoting the number of queries.
- Each of the next q lines contains a single non-negative integer k ($0 \leq k \leq 10^9$), denoting you must compute $k\text{-mex}(a)$ for this k .

It is guaranteed that $\sum n \leq 5 \times 10^3$ and $\sum q \leq 5 \times 10^5$ across all test cases in one test file.

Output

For each test case, output a single integer, denoting the bitwise XOR (exclusive-OR) of the answers to all queries in that test case.

Examples

Sample Input 1	Sample Output 1
2 3 0 1 3 3 1 2 5 9 0 0 1 1 2 3 3 4 5 7 4 5 6 7 8 9 10	3 8